Bending an ice shelf with meltwater

A 2-D viscoelastic beam model: derivation and numerics

3 November 2026

This note derives, from scratch, a simple model of how a floating ice shelf bends when a meltwater lake fills (and drains) on its surface, and then shows how to solve it on a computer in a few lines. Every step comes with a picture. The code is in `flexure/viscoelastic_beam.py`. The figures come from `flexure/make_diagrams.py` and `flexure/run_examples.py`.

The whole model in two lines. The shelf is a thin Maxwell (spring-and-dashpot) beam floating on sea water. The deflection $w(x,t)$, positive downwards, obeys

$$\left(D\,\frac{\partial^4}{\partial x^4} + k\right)\frac{\partial w}{\partial t} \;=\; \frac{\partial q}{\partial t} \;+\; \frac{q - k\,w}{\tau},$$

where $q = \rho_w g\, d$ is the weight of a lake of depth d, $k = \rho_{sw} g$ is the stiffness of buoyancy, D is the flexural rigidity and τ is the Maxwell relaxation time. When a lake first appears, the ice responds **elastically**. Over a few days it relaxes towards **local isostasy**, $w = q/k$.

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1 Set-up and assumptions

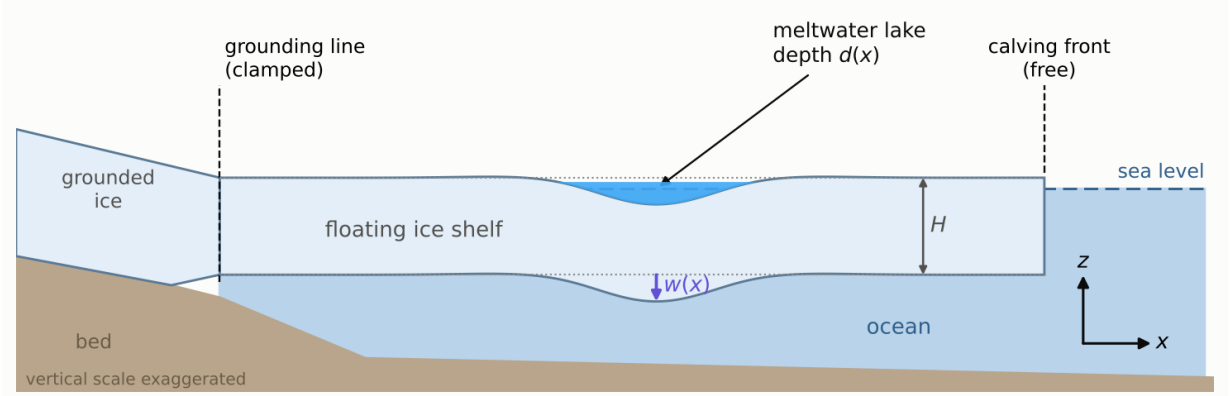


Figure 1: A vertical slice through the problem. A meltwater lake of depth $d(x)$ sits on a floating ice shelf of thickness H, which deflects downwards by $w(x)$. Vertical scale exaggerated.

We take a vertical slice through the shelf along a line x that runs from the grounding line towards the calving front. z points up. The shelf has thickness H, floats in sea water, and carries a lake of water depth $d(x,t)$. The lake's weight pushes the shelf down by $w(x,t)$, measured **positive downwards**. The change in surface elevation is therefore $-w$.

We assume the following:

assumption	why it is reasonable
2-D, plane strain. Nothing varies along the third direction y, so the lake is a long trough.	It keeps the maths one-dimensional. A circular lake gives similar numbers (see §13).
Thin beam. Plane cross-sections stay plane.	Lakes are 1 to 5 km across and $H \approx 100$ to 500 m.
Small deflections. $w \ll H$, small slopes.	w is a few metres.
Linear Maxwell rheology.	This is the simplest material that is elastic over hours and viscous over weeks.
Hydrostatic ocean. Sea water pushes back like a bed of springs.	Lakes fill and drain over hours to weeks, much slower than ocean waves.
Prescribed load $q = \rho_w g d(x,t)$.	We specify the water and do not route it (see §13).
No stretching or flow of the shelf.	Bending happens much faster than the shelf spreads.

Everything is linear in the **perturbation**. The background weight of the ice is already balanced by the background buoyancy, so only the extra load q and the extra deflection w appear.

2 Buoyancy is a bed of springs

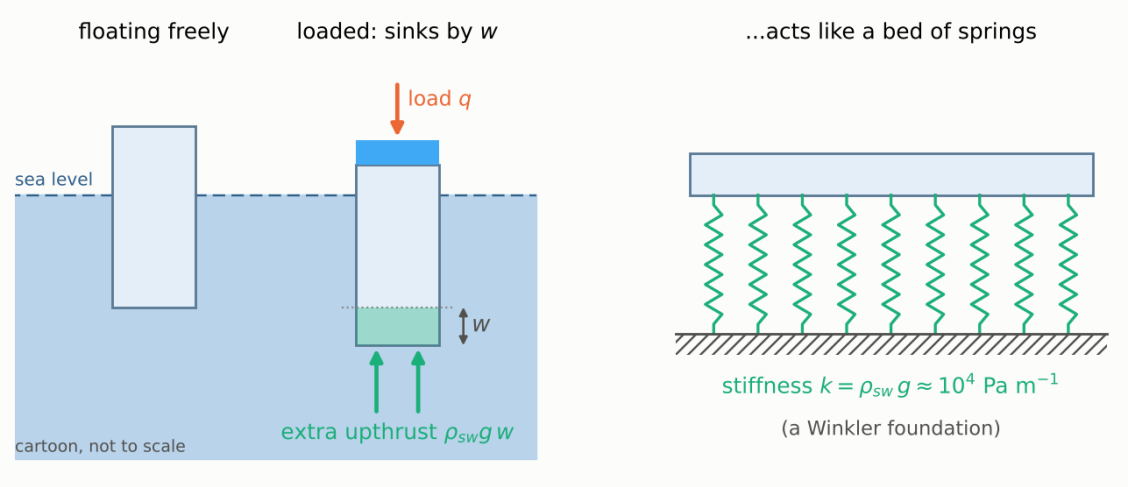


Figure 2: Pushing a floating column down by w displaces extra sea water and produces an extra upthrust $\rho_{sw} g\, w$. Buoyancy therefore acts like a bed of springs of stiffness $k = \rho_{sw} g$.

Push a floating column down by w and it displaces an extra w of sea water per unit area. Archimedes then gives an extra upward pressure

$$p_{\text{buoy}} = \rho_{sw}\, g\, w \;\equiv\; k\, w, \qquad k = \rho_{sw} g \approx 1.0 \times 10^4 \text{ Pa m}^{-1}.$$

This acts exactly like a bed of independent springs with stiffness k, which is called a *Winkler foundation*. The spring is the ocean, not the ice.

A useful by-product: if nothing else resisted, a lake of depth d would sink the ice by

$$w_{\text{iso}} = \frac{q}{k} = \frac{\rho_w}{\rho_{sw}}\, d \approx 0.97\, d.$$

3 Force balance on a slice

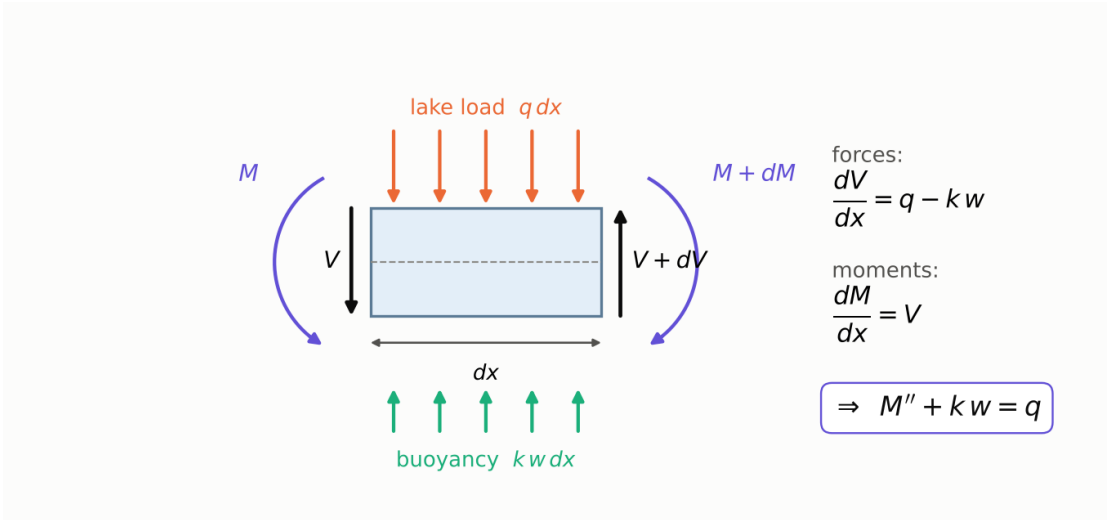


Figure 3: Forces and moments on a slice of the shelf of width dx.

Cut out a slice of width dx. Four things act on it:

- the lake load $q\,dx$, pushing down;
- buoyancy $k\,w\,dx$, pushing up;
- shear forces V and $V + dV$ on its two faces, which come from the neighbouring ice;
- bending moments M and $M + dM$ on its two faces.

Vertical forces must balance: $dV = (q - kw)\,dx$. **Moments** must balance (taken about the left face, ignoring tiny dx^2 terms): $dM = V\,dx$. Eliminating V gives the equilibrium equation

$$\boxed{\frac{\partial^2 M}{\partial x^2} + k\,w = q} \tag{1}$$

In words: wherever the lake is not held up by buoyancy, the ice must carry the difference through changes in its bending moment.

4 Bending: curvature, strain and moment

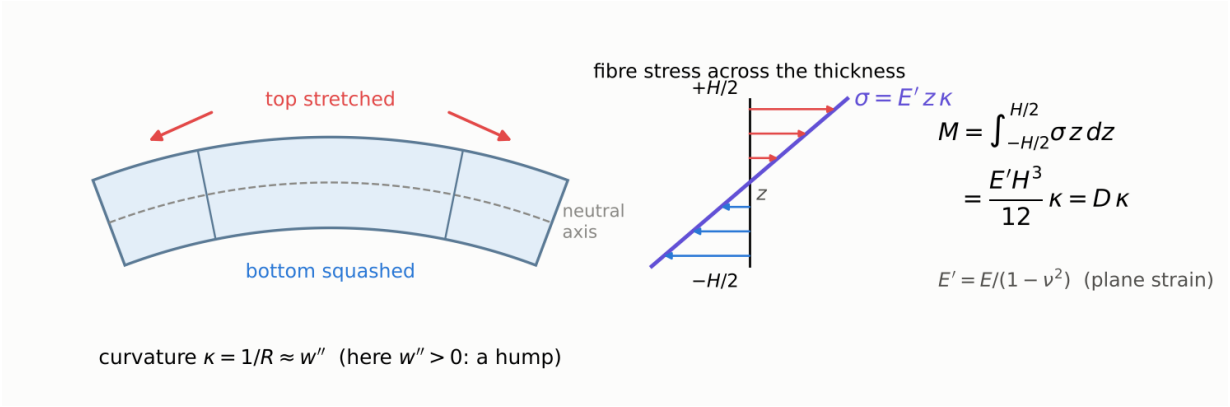


Figure 4: Cross-sections stay plane as the beam bends, so fibre strain and stress vary linearly with height z. Integrating σz across the thickness gives $M = D\kappa$.

When the beam bends, cross-sections stay straight but tilt. A fibre at height z above the mid-plane (the *neutral axis*) is stretched in proportion to its distance from it:

$$\varepsilon(z) = z\,\kappa, \qquad \kappa = \frac{\partial^2 w}{\partial x^2} \quad \text{(the curvature)}.$$

With w positive down, $\kappa > 0$ means a hump. The top is stretched (tension) and the bottom squashed. Under the middle of a lake the ice sags, so $\kappa < 0$ and the top is in compression.

The **bending moment** is the turning effect of all the fibre stresses $\sigma(z)$:

$$M = \int_{-H/2}^{H/2} \sigma(z)\, z\, dz.$$

For an **elastic** beam in plane strain, $\sigma = E'\varepsilon$ with $E' = E/(1-\nu^2)$. So

$$M = E'\kappa \int_{-H/2}^{H/2} z^2\, dz = \frac{E' H^3}{12}\,\kappa \equiv D\,\kappa, \qquad D = \frac{E H^3}{12(1-\nu^2)}.$$

D is the **flexural rigidity**. It scales as H^3, so a shelf twice as thick is eight times as stiff.

For a **viscous** beam, stress depends on strain *rate*. For an incompressible Newtonian fluid of viscosity η in plane strain with $\sigma_{zz} \approx 0$, this is $\sigma = 4\eta\,\dot{\varepsilon}$. The same integral gives

$$M = \frac{4\eta H^3}{12}\,\dot{\kappa} \equiv \Lambda\,\dot{\kappa}, \qquad \Lambda = \frac{\eta H^3}{3}.$$

Λ is a "viscous flexural rigidity".

5 Ice is a Maxwell material

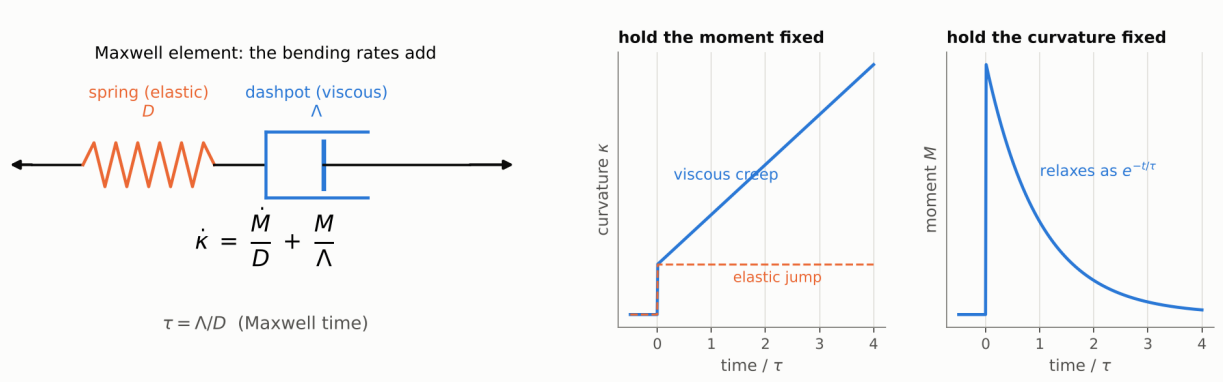


Figure 5: A Maxwell element (a spring and a dashpot in series) and its two classic responses: creep under a fixed moment, and relaxation at fixed curvature.

Over minutes to hours, ice behaves like a spring. Over days to weeks it creeps like a very stiff fluid. The simplest model that does both is a **Maxwell element**, a spring and a dashpot in series. Each fibre obeys $\dot{\varepsilon} = \dot{\sigma}/E' + \sigma/4\eta$. Multiply by z and integrate across the thickness, exactly as above, and the *bending* rates simply add:

$$\boxed{\dot{\kappa} = \frac{\dot{M}}{D} + \frac{M}{\Lambda}} \qquad \Longleftrightarrow \qquad \dot{M} = D\,\dot{\kappa} - \frac{M}{\tau}, \tag{2}$$

with the **Maxwell time**

$$\tau = \frac{\Lambda}{D} = \frac{4\eta(1-\nu^2)}{E}.$$

For $\nu \to \frac{1}{2}$ this becomes the familiar η/G, where G is the shear modulus. With $E = 1$ GPa and $\eta = 10^{14}$ Pa s, $\tau \approx 4.2$ days. The right-hand plots show the two classic experiments:

- **Hold M fixed.** κ jumps elastically and then creeps linearly.
- **Hold κ fixed.** M jumps and then relaxes as $e^{-t/\tau}$.

A useful mental picture is an **elastic beam whose stress-free shape slowly drifts towards whatever shape it is being held in**.

6 The master equation

Now combine the force balance (1) and the material law (2). Take $\partial^2/\partial x^2$ of (2), use (1) to replace M'' by $q - kw$, and use (1) differentiated in time, $\dot{M}'' = \dot{q} - k\dot{w}$:

$$\dot{q} - k\,\dot{w} \;=\; D\,\frac{\partial^4 \dot{w}}{\partial x^4} \;-\; \frac{q - kw}{\tau}$$

Rearranged:

$$\boxed{\left(D\,\frac{\partial^4}{\partial x^4} + k\right)\frac{\partial w}{\partial t} = \frac{\partial q}{\partial t} + \frac{q - k\,w}{\tau}} \tag{3}$$

Read it left to right:

- The operator in brackets is the **elastic** beam-on-springs operator.
- $\partial q/\partial t$ is the **instantaneous elastic** response to any change in load.
- $(q - kw)/\tau$ is the **creep** term. Wherever buoyancy is not yet carrying the lake ($kw \neq q$), the ice creeps at a rate set by τ, and it stops only when $kw = q$.

7 Two limits: elastic and isostatic

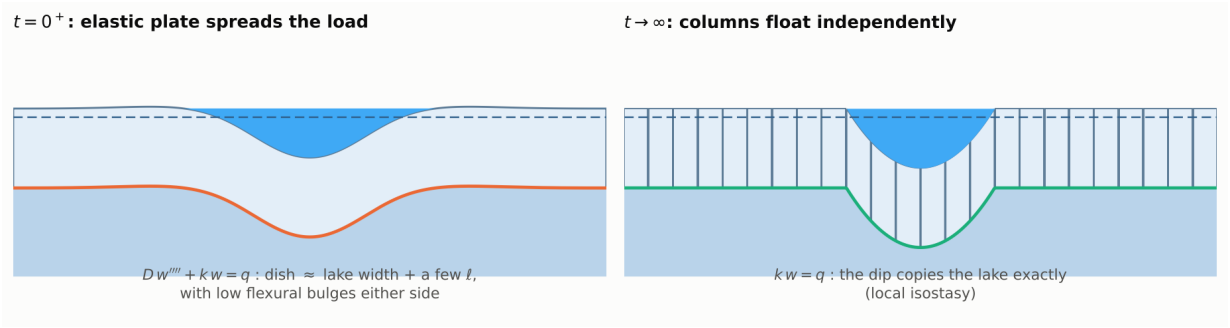


Figure 6: The two limits of the model: an elastic response at $t = 0^+$ and local isostasy as $t \to \infty$.

At $t = 0^+$ the response is elastic. Integrate (3) across the instant the load appears. The creep term is finite, so it contributes nothing in zero time, and we are left with

$$D\,\frac{\partial^4 w}{\partial x^4} + k\,w = q.$$

This is the classic "beam on elastic foundation" (Hetényi, 1946). Its natural length scale is the **flexural length**

$$\ell = \left(\frac{D}{k}\right)^{1/4} \approx 610 \text{ m} \quad (H = 250 \text{ m},\ E = 1 \text{ GPa}).$$

For a line load P the solution is $w = \frac{P\beta}{2k}\,e^{-\beta|x|}\big(\cos\beta|x| + \sin\beta|x|\big)$, with $\beta = 1/(\sqrt{2}\,\ell)$.

- The dip is spread over a few flexural lengths.
- It crosses zero at $|x| = 3\pi\ell/(2\sqrt{2}) \approx 3.3\ell$.

- A low **flexural bulge** peaks at $|x| = \sqrt{2}\,\pi\,\ell \approx 4.4\ell$, with height $e^{-\pi} \approx 4\%$ of the central dip.

Our 3 km-wide, 4 m-deep lake sinks the centre by 3.2 m at first, instead of 3.9 m.

As $t \to \infty$ the response is isostatic. The left side of (3) goes to zero when nothing changes any more, so $q = kw$. The ice "forgets" its stiffness and floats as independent columns. The dip copies the lake exactly, and the bending moment relaxes to zero.

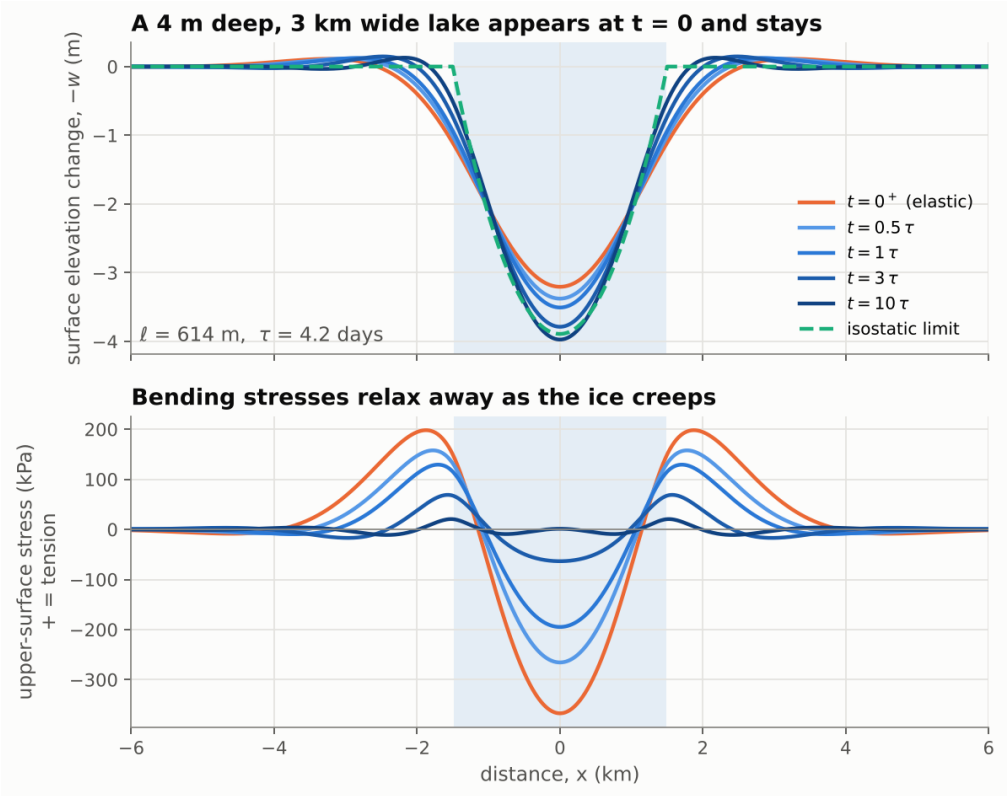


Figure 7: A lake appears at $t = 0$ and is left there. The orange curve is the elastic response. The blue curves (light to dark) show the ice creeping towards the dashed isostatic shape. The bending stresses (bottom panel) decay as it does.

8 Which shapes relax first? The relaxation spectrum

Because (3) is linear with constant coefficients, each sinusoid in x evolves on its own. Put $w = \hat{w}(t)\cos k_x x$ and $q = \hat{q}\cos k_x x$, with $\hat{q}$ switched on at $t = 0$. Then (3) becomes an ordinary differential equation with the solution

$$\hat{w}(t) = \underbrace{\frac{\hat{q}}{k}}_{\text{isostatic}} + \Big(\underbrace{\frac{\hat{q}}{Dk_x^4 + k}}_{\text{elastic}} - \frac{\hat{q}}{k}\Big)\, e^{-t/\tau_\lambda}, \qquad \boxed{\tau_\lambda = \tau\big[1 + (k_x\ell)^4\big] = \tau\Big[1 + \Big(\frac{2\pi\ell}{\lambda}\Big)^4\Big]}$$

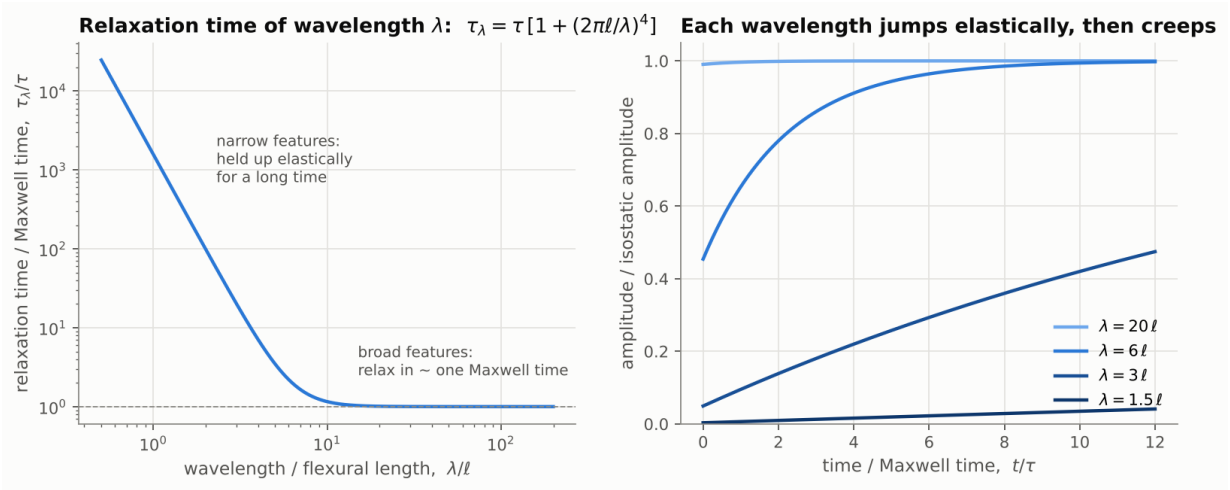


Figure 8: Left: relaxation time against wavelength. Right: each wavelength jumps elastically and then relaxes towards isostasy at its own rate.

- **Broad features** ($\lambda \gg 2\pi\ell$) are barely supported by bending in the first place. They are already nearly isostatic, and what is left relaxes in about one Maxwell time.
- **Narrow features** ($\lambda \lesssim 2\pi\ell$) are held up mainly by the ice's stiffness. They relax extremely slowly, as $(2\pi\ell/\lambda)^4$. On a 250 m-thick shelf, a 2 km-wavelength wiggle takes about two months to relax and a 1 km one takes more than two years.

This also tells us that the problem is **not stiff** numerically. The fastest decay rate of any mode is $1/\tau$, so a time step of a fraction of τ is enough for every wavelength.

9 Stresses, and why draining is dangerous

The bending stress is largest at the upper surface, where crevasses open:

$$\sigma_{\text{top}} = \frac{6M}{H^2} \qquad (\text{positive} = \text{tension}).$$

For the 4 m-deep, 3 km-wide lake:

- The elastic response puts the surface under the lake centre into compression of about -370 kPa.
- A ring of **tension, about** $+200$ **kPa**, sits just outside the lake shore.
- These stresses are of the same order as those usually needed to open surface crevasses, roughly 100 kPa or more (e.g. Vaughan, 1993).

This is the "ring fracture" mechanism proposed for the 2002 Larsen B break-up (Banwell et al., 2013).

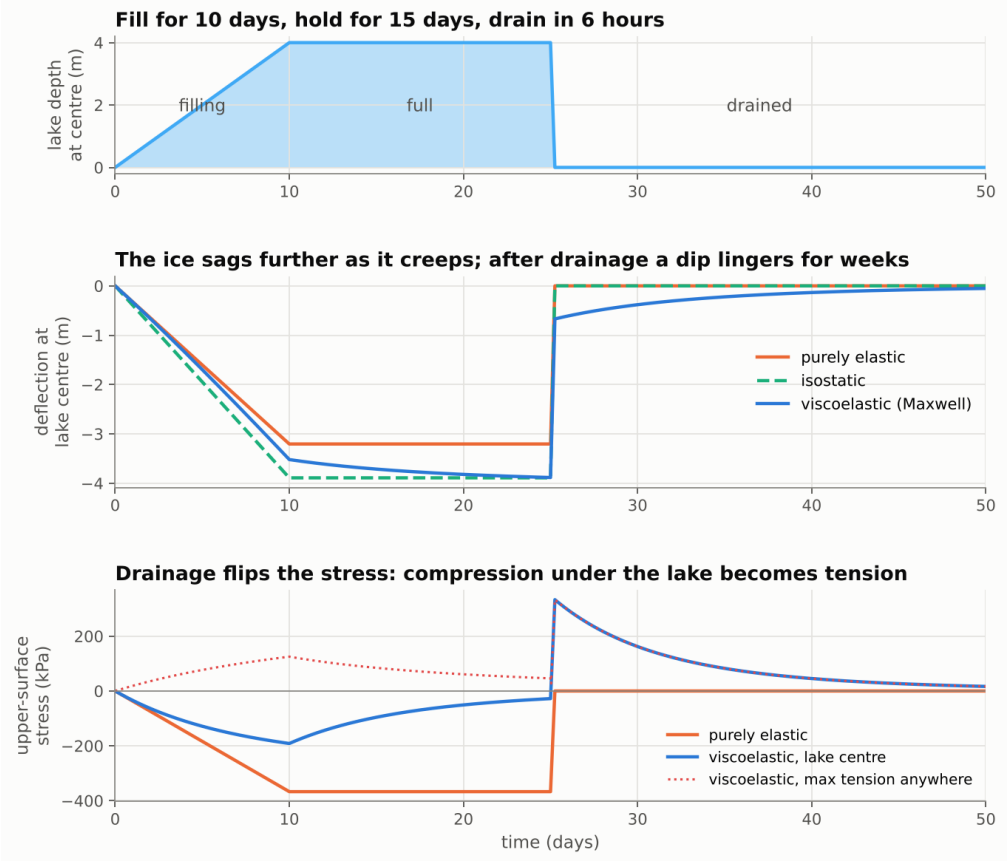


Figure 9: A lake fills for 10 days, is held for 15 days and drains in 6 hours. Top: lake depth. Middle: deflection at the lake centre. Bottom: bending stress at the upper surface.

The viscoelastic twist comes with **drainage**:

- **While the lake sits**, the moments relax ($M \to 0$). The ice "gets used to" being bent.
- **When the lake drains** in hours (for example by hydrofracture), the load vanishes much faster than τ. The response to *removing* the load is again elastic, and equal and opposite to the original elastic response. Superposed on a relaxed state, this leaves the ice **bent upwards**, with the surface under the former lake now in **tension (+330 kPa)**.
- A dip of about 0.7 m lingers at the old lake site and creeps away over the following weeks.

Repeated fill-drain cycles therefore reverse the stress at the lake centre again and again. This fatigue-like loading is what the model was built to explore (compare MacAyeal & Sergienko, 2013; Banwell et al., 2019).

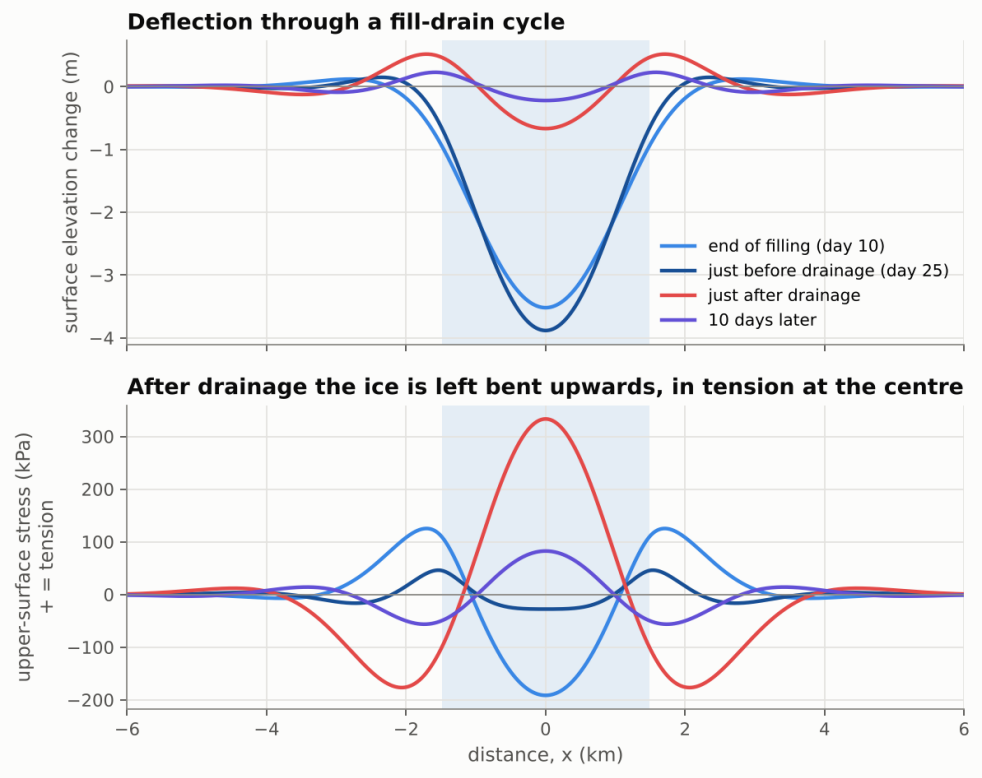


Figure 10: Deflection and upper-surface stress at key moments of the fill-drain cycle.

10 Boundary conditions

The beam equation is fourth order, so each end needs **two** conditions.

end	physical meaning	conditions	type
grounding line	ice frozen to the bed: no deflection, no tilt	$w = 0,\ \partial w/\partial x = 0$ ("clamped")	kinematic: on w
calving front	nothing holds the end	$M = 0,\ V = \partial M/\partial x = 0$ ("free")	dynamic: on M

The free-end conditions are on M, not directly on w. For a Maxwell beam that starts from rest, $M = 0$ for all time implies $\partial^2 w/\partial x^2 = 0$ for all time, and similarly for the third derivative, so the elastic intuition carries over.

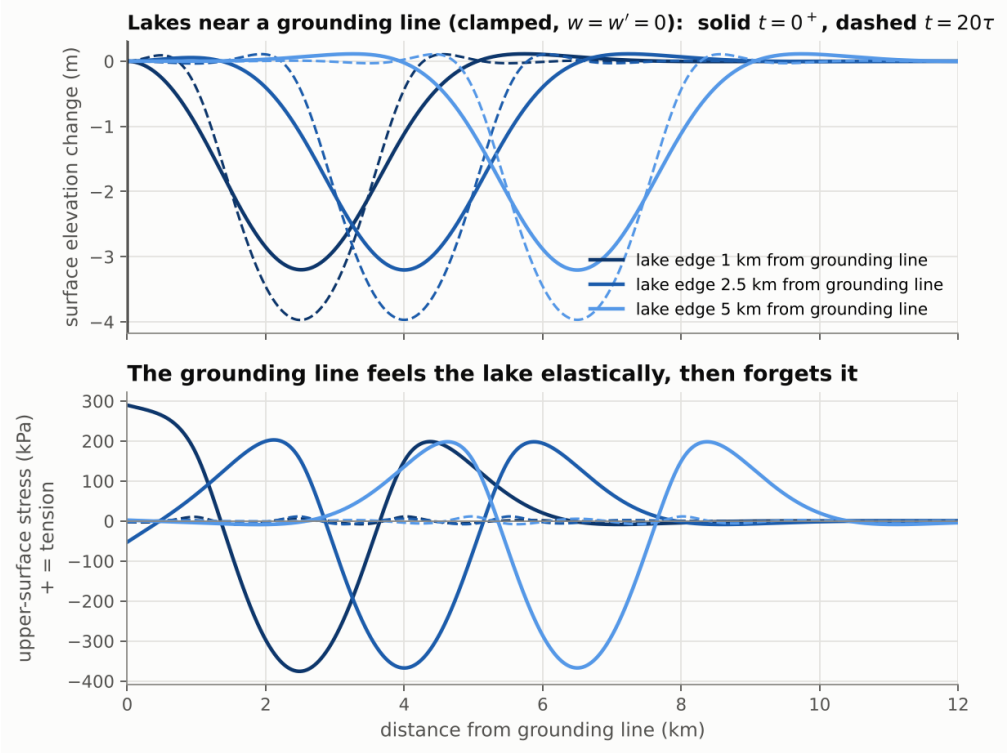


Figure 11: Lakes 1, 2.5 and 5 km from a clamped grounding line. A lake within about 2ℓ of the grounding line strains it elastically: here, +290 kPa of surface tension at the grounding line for the closest lake. That stress relaxes away (dashed curves) as the shelf creeps.

11 Numerics: fast and simple

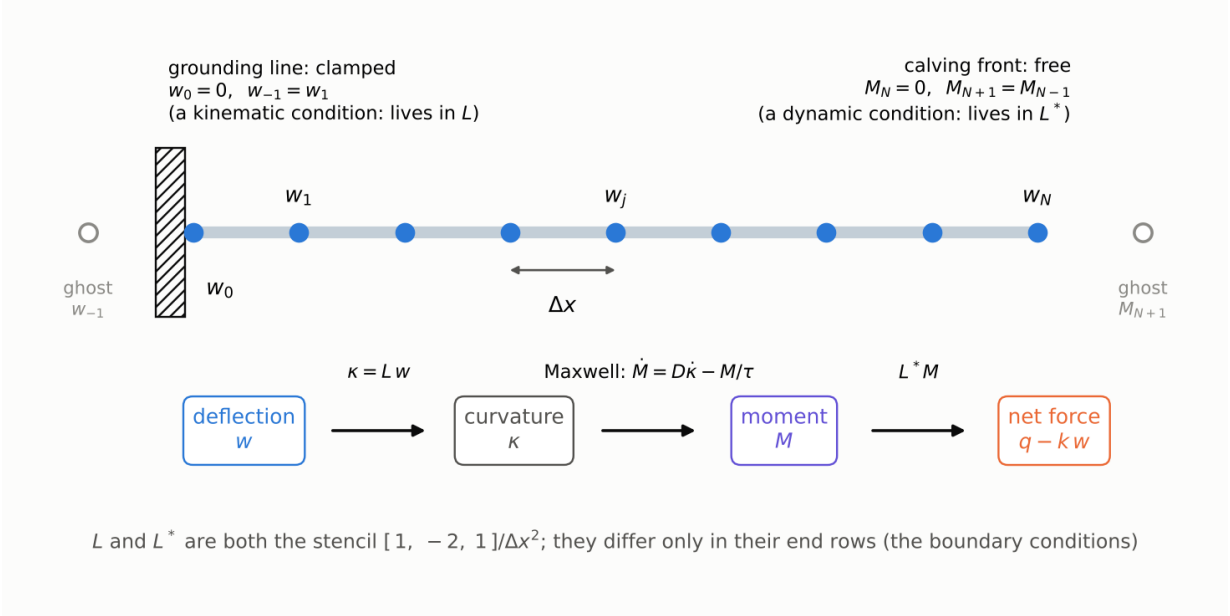


Figure 12: The grid, its ghost nodes and the chain of operators $w \to \kappa \to M \to$ force. Kinematic boundary conditions live in L and dynamic ones in L^*.

11.1 Space: two little matrices

Put N equally spaced nodes x_j along the beam, with spacing Δx, and store w_j and M_j at each node. The physics is a chain of three steps:

$$w \xrightarrow{L} \kappa \xrightarrow{\text{Maxwell}} M \xrightarrow{L^*} \text{net force } (q - kw).$$

Both L and L^* are the familiar second-difference stencil $[1, -2, 1]/\Delta x^2$. They differ **only in their end rows**, which is where the boundary conditions live:

end type	row of L (curvature from w)	row of L^* (force from M)
clamped (node 0)	ghost $w_{-1} = w_1$: $\kappa_0 = (2w_1 - 2w_0)/\Delta x^2$	not used: the equation is replaced by $w_0 = 0$
free (node 0)	not used: zeros, so $M_0 \equiv 0$	ghost $M_{-1} = M_1$: $f_0 = (2M_1 - 2M_0)/\Delta x^2$

Kinematic conditions go in L and dynamic ones go in L^*. The fourth derivative is never written down explicitly. It is just $L^* D\, L$, so a spatially varying $D(x)$ (a thinning shelf) costs nothing extra.

11.2 Time: the trapezoidal rule, then one linear solve

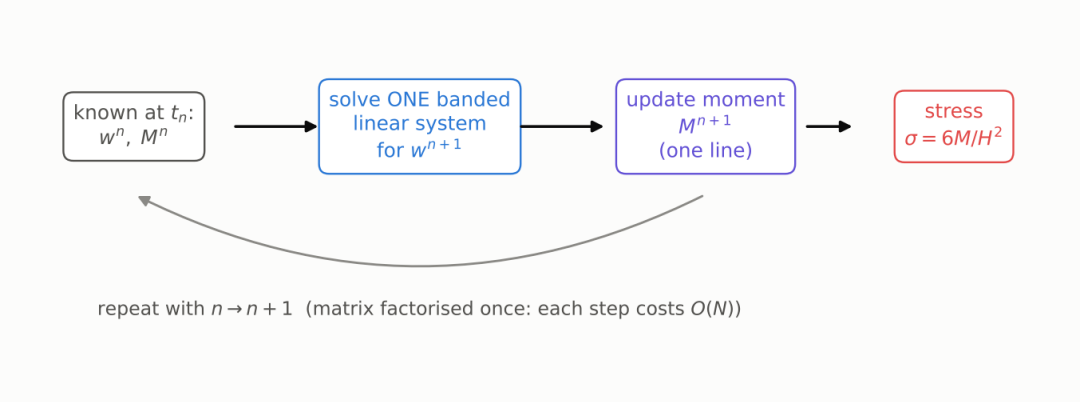


Figure 13: One time step of the algorithm.

Step from t_n to $t_{n+1} = t_n + \Delta t$. Write $a = \Delta t/\tau$.

1. Moment update. Apply the trapezoidal (Crank–Nicolson) rule to the Maxwell law $\dot{M} = DL\dot{w} - M/\tau$ at every node:

$$M^{n+1} = c\left[(1 - \tfrac{a}{2})\, M^n + D\, L\, (w^{n+1} - w^n)\right], \qquad c = \frac{1}{1 + a/2}.$$

2. Deflection solve. Substitute this into the force balance at the new time, $L^* M^{n+1} + k\, w^{n+1} = q^{n+1}$. This gives **one linear system** for the new deflection:

$$\Big(\underbrace{L^* cD\, L + k\, I}_{\text{constant, pentadiagonal}}\Big)\, w^{n+1} = q^{n+1} + L^* cD\, L\, w^n - L^*\left[c\,(1 - \tfrac{a}{2})\, M^n\right].$$

3. Repeat. Solve step 2 for w^{n+1}, then evaluate step 1 for M^{n+1}.

That is the entire algorithm. In the code it reads:

```
def step(self, w0, M0, q1, dt):
    lu, S, c, a = self._step_matrices(dt)          # S = L* (c D) L, factorised once
    rhs = q1 + S @ w0 - self.Lstar @ (c * (1 - a/2) * M0)
    w1 = self._solve(lu, rhs)                       # one banded back-substitution
    M1 = c * ((1 - a/2) * M0 + self.D * (self.L @ (w1 - w0)))
    return w1, M1
```

11.3 Why this is a good scheme

- **Fast.** The matrix never changes, so it is factorised once. Each step is then one banded back-substitution costing $O(N)$. 3000 nodes × 1200 steps take about one second in pure Python and SciPy.
- **Exact elastic limit.** With $\Delta t/\tau \to 0$ it reduces to the elastic solve, so instantaneous loading and unloading are captured exactly.
- **Second-order accurate** in both Δx and Δt, and **unconditionally stable**: the trapezoidal rule is A-stable.
- **Force balance holds to round-off at every step.** The two operators are used consistently, so the discrete version of (1) is satisfied exactly, and the stresses are physically consistent.
- **Choosing steps.** $\Delta x \lesssim \ell/10$ and $\Delta t \lesssim \tau/10$ give errors well below 0.1 % (see the convergence panel below).

12 Verification

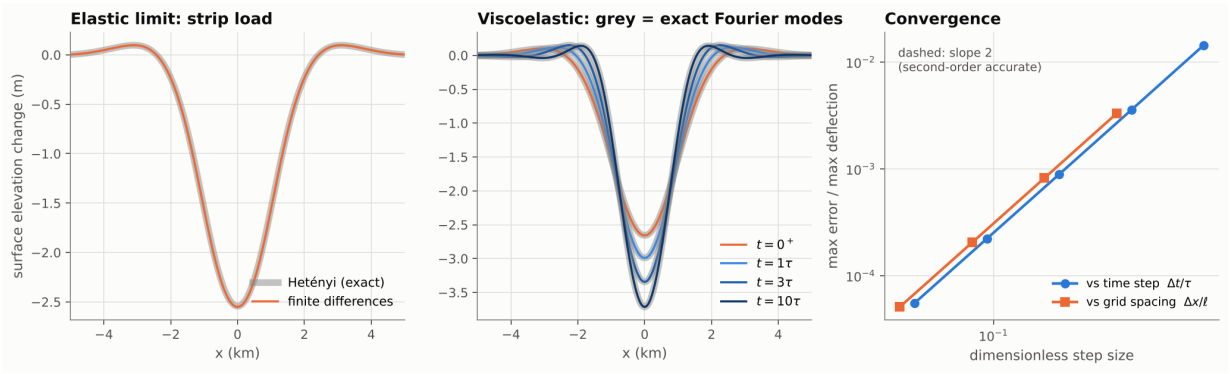


Figure 14: Verification. Left: elastic strip load against Hetényi's exact solution. Centre: viscoelastic relaxation against the exact Fourier-mode solution (grey). Right: second-order convergence in Δt and Δx.

The test-suite (`flexure/tests/test_viscoelastic_beam.py`, run with `python -m pytest -q`) checks the model against exact solutions:

test	exact solution	agreement
elastic strip load, free beam	Hetényi (1946)	< 0.2 %
line load at a free calving front	$w = \frac{2P\beta}{k} e^{-\beta x}\cos\beta x$	< 0.2 %
uniform load, clamped grounding line	$w = \frac{q}{k}\left[1 - e^{-\beta x}(\cos\beta x + \sin\beta x)\right]$ (the tidal-flexure profile)	< 0.1 %

test	exact solution	agreement
viscoelastic relaxation of a lake	exact Fourier-mode solution (§8)	< 0.2 %
long-time limit	local isostasy $w = q/k$, $M \to 0$	✓
uniform load on a free beam	no bending at all	to round-off
discrete force balance $L^*M + kw = q$	(all boundary-condition combinations)	to round-off
time convergence	slope 2	✓

13 Limitations and easy extensions

- **Lake–flexure feedback.** Here the water depth is prescribed. In reality the lake sits in the depression it creates. Coupling the model to the fill-spill-merge routing in `python` would let water and ice shape each other.
- **Variable thickness.** This already works: pass `H` as an array. Both D and Λ scale as H^3, so τ is unchanged.
- **3-D lakes.** For a circular lake the same physics gives a plate equation in polar coordinates, with Bessel-function solutions. The ring of tension survives, with similar magnitudes.
- **Nonlinear (Glen's law) creep and depth-varying temperature.** Here η depends on stress and depth, so Λ becomes a function of the state. The moment update becomes a local nonlinear update; the solve stays the same.
- **Membrane stresses and ice-shelf flow.** These are neglected, which is fine for bending over days to weeks. For coupled flow and flexure see MacAyeal et al. (2021).
- **Fracture.** The model reports stresses but does not break.

14 Parameters and references

symbol	meaning	default
H	ice thickness	250 m
E	effective Young's modulus (tidal-flexure studies infer about 1 GPa, well below the laboratory 9 GPa; e.g. Vaughan, 1995)	1 GPa
ν	Poisson's ratio	0.3
η	ice viscosity	10^{14} Pa s
ρ_i, ρ_{sw}, ρ_w	ice, sea water, meltwater densities	917, 1028, 1000 kg m^{-3}
$D = EH^3/12(1-\nu^2)$	flexural rigidity	1.4×10^{15} N m
$k = \rho_{sw} g$	buoyancy stiffness	1.0×10^4 Pa m^{-1}
$\ell = (D/k)^{1/4}$	flexural length	614 m
$\tau = 4\eta(1-\nu^2)/E$	Maxwell time	4.2 days

The values of E and especially η are uncertain by an order of magnitude or more. Treat the numbers above as illustrative. Because $\ell \propto E^{1/4} H^{3/4}$ and $\tau \propto \eta/E$, it is easy to rescale the results.

- Banwell, A. F., MacAyeal, D. R., & Sergienko, O. V. (2013). Breakup of the Larsen B Ice Shelf triggered by chain reaction drainage of supraglacial lakes. *Geophysical Research Letters*, 40, 5872–5876.

- Banwell, A. F., Willis, I. C., Macdonald, G. J., Goodsell, B., & MacAyeal, D. R. (2019). Direct measurements of ice-shelf flexure caused by surface meltwater ponding and drainage. *Nature Communications*, 10, 730.
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